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Lindsay R.L. Larson, Jacqueline K. Eastman & Dora E. Bock

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A MULTI-METHOD EXPLORATION OF THE RELATIONSHIP BETWEEN KNOWLEDGE AND RISK: THE IMPACT ON MILLENNIALS' RETIREMENT INVESTMENT DECISIONS

Lindsay R.L. Larson, Jacqueline K. Eastman^{ORCID}, and Dora E. Bock^{ORCID}

The millennial cohort has faced a unique environment that may have a lasting impact on the financial investment decisions they make as adults. A multimethod set of studies investigates how knowledge and risk interact to inform millennial retirement investment choices. Study 1 suggests a decline in risk-taking for those with low confidence in their financial knowledge. Study 2 reveals that low financial literacy enhances susceptibility to the influence of “feelings as information” when making retirement decisions. Study 3 utilizes qualitative data to consider the connection between millennial financial decision making and the climate of risk brought on by the Great Recession.

Retirement investment decisions are among the most important decisions a consumer makes (Kozup, Howlett, and Pagano 2008), yet years of research on financial decision making have illustrated that consumers tend to make suboptimal financial choices (Vlaev, Chater, and Stewart 2009). Recent economic turmoil raises the importance of this problem (Lusardi and Tufano 2009), and set against the increasing complexity in financial products (Lusardi 2012; Lusardi and Mitchell 2007), these issues highlight the importance of understanding factors that guide consumer investment decisions. There is particular pressure on young consumers who often have high levels of debt and low levels of financial knowledge (Lusardi, Mitchell, and Curto 2010), to make sound investment decisions. Young consumers entering the workforce are presented with difficult financial decisions, and financial mistakes made at this time in life can be costly (Lusardi, Mitchell, and Curto 2010) as these consumers may not be able to rely on employer pensions or Social Security to fund retirement (Lusardi 2012). As the

overall investment market is estimated to have \$1.2 trillion in annual sales with younger investors making up approximately 15 percent of that market in the United States (Cudmore et al. 2010), and given concerns that today's youth will be financially unprepared for retirement (Kelly 2013; Olson 2014), an understanding of this segment is of great significance. The current set of studies examines how college-age millennial consumers who are about to enter the workforce make retirement investment decisions.

There are considerable consumer implications surrounding retirement investment decisions, as bad investment decisions can negatively impact quality of life when it comes time to retire and may increase the likelihood of reliance on government assistance programs (Kozup, Howlett, and Pagano 2008). With a decline in employer-sponsored defined benefit pension plans and a rise in financial solvency problems with Social Security, individuals must take on greater responsibility towards securing their own financial wellbeing (Lusardi 2012). In terms of preparing for retirement, there is now a high reliance on individual retirement planning (i.e., utilizing defined contribution plans and individual retirement accounts). These plans started with the Baby Boom generation (Lusardi 2012) and will continue as the pension burden and risk is shifted from corporations to employees (Kozup, Howlett, and Pagano 2008; Mittelstaedt 2004). With increasing pressure on consumers to take charge of their retirement investment planning, research is needed to examine some of the key variables known to impact retirement investment decisions, such as financial knowledge and risk tolerance (Eccles et al.

Lindsay R.L. Larson (Ph.D., Yale University), Assistant Professor of Marketing, College of Business Administration, Georgia Southern University, Statesboro, GA, lindsaylarson@georgiasouthern.edu.

Jacqueline K. Eastman (Ph.D., Florida State University), Professor of Marketing, College of Business Administration, Georgia Southern University, Statesboro, GA, jeastman@georgiasouthern.edu.

Dora E. Bock (Ph.D., Louisiana State University), Assistant Professor of Marketing, Raymond J. Harbert College of Business, Auburn University, Auburn, AL, deb0022@auburn.edu.

2013). This connection between knowledge, risk, and investment is especially relevant to the millennial cohort, considering that high risk aversion (Debevec et al. 2013) and low levels of financial knowledge (Lusardi, Mitchell, and Curto 2010) are recognized as defining attributes of that generation.

Generational cohort theory (Strauss and Howe 1991) suggests that one way of understanding consumer differences is through looking at generational cohorts: groups of people born during a specific time period who share the same historic environment and life experiences. These differences can impact generational tastes and preferences, such as attitudes toward saving (Meredith and Schewe 1994), finance, and insurance products (Schewe, Meredith, and Noble 2000). Macroeconomic events during an individual's early formative years can actually provoke long-term effects upon their financial risk attitudes, even when controlling for age, year effects, and household characteristics (Malmendier and Nagel 2011). Millennials have witnessed significant traumatic events including 9/11, the War on Terrorism, and mass shootings such as Columbine (Morton 2002) during their formative years. In terms of financial events, this cohort has seen tech bubbles, housing bubbles, stock market crashes, Wall Street scandals, and the Great Recession, as well as notable periods of high unemployment and declining salaries (Davis 2012; Morton 2002). The life course approach (Elder 2006) to studying behavioral patterns impacting generational cohorts supports the notion that activities such as retirement planning may be directly influenced by the socioeconomic, historical, and political context to which one is exposed during key developmental years. The millennial cohort came of age "during the greatest economic disaster since the Great Depression" (Stanley 2013, p. 37), which may have significantly impacted the mindset of this cohort when it comes to taking risks. Existing research also notes the importance of studying individuals as they first begin to build their lifelong investment preferences and tendencies, which is typically when people are in their early twenties (Cayla, Beers, and Arnould 2014).

Research finds clear differences in how millennials view investment decisions compared to other generations. Millennials entering adulthood are known to be financially risk averse and conservative investors, who more often than previous generations keep their savings in cash and less in stocks (Debevec et al. 2013; Insured Retirement Institute 2013; Sharf 2014). In fact, research

examining cohort differences reveals that millennials maintain notably more conservative financial tendencies than baby boomers (Schewe, Meredith, and Noble 2000). Another cohort difference regards the pursuit of financial assistance. Zick, Mayer, and Glaubitz (2012) compare cohort differences in response to the Great Recession and find that older cohorts are more likely to seek the assistance of a professional financial advisor than millennials; yet college-age millennials remain optimistic that they will have sufficient funds for retirement (De Hauw and De Vos 2010). This optimism may be due to a lack of maturity (Debevec et al. 2013) and may be misplaced, as millennials in the workforce often fail to participate in employer retirement plans or contribute at a sufficient rate, and/or maintain appropriate levels of diversification (Kelly 2013). This issue is likely compounded by the fact that millennials tend to exhibit lower levels of financial knowledge and higher levels of risk aversion than previous generations (Debevec et al. 2013; Lusardi, Mitchell, and Curto 2010).

In finance, a basic principle is that investment return is dependent on the level of risk involved with the investment, with greater risk requiring greater return (Ross, Westerfield, and Jordan 2011). If consumers are too risk averse, they may not take the level of risk needed with their investment strategy to be financially prepared for retirement. Mistakes made with investment diversification could reduce consumer retirement income by one-fifth (Olson 2014). For millennials, there is particular pressure and financial anxiety as they face notably high levels of student loan debt and decreased job opportunities in today's economy (Davis 2012; Lusardi, Mitchell, and Curto 2010). It is, therefore, vital that millennials start to consider their investments seriously to ensure that they will be financially prepared for retirement (Olson 2014).

The sheer size and financial worth of the youth market indicates the importance of millennials to marketers (Cudmore et al. 2010). The size of this cohort, estimated to be up to 92 million consumers (Stanley 2013), suggests that the financial mistakes made by millennials today have significant long-term implications (Davis 2012; Lusardi, Mitchell, and Curto 2010). It is predicted that millennials will need at least twice the amount of money to retire than current workers nearing retirement age (Gillespie 2011). For financial managers, millennials represent a tremendous market as they control an estimated two trillion dollars in liquid assets currently, which is forecasted to increase to seven trillion dollars by the end

of the decade (Sharf 2014). Understanding the millennial investor represents an opportunity to financial services marketers.

The purpose of this research is to understand millennial investment decision making better, with particular focus on how financial knowledge (i.e., an individual's understanding of personal finance) might buffer the impact of today's risky climate upon one's preparation for the future. Through a multimethod series of three studies—a survey, an experiment, and a qualitative investigation—we aim to highlight the related impact of financial knowledge and risk on retirement investment decisions of millennials. Study 1 investigates how millennial retirement investment decisions are influenced by perceived knowledge and risk tolerance. Study 2 assesses whether the influence of a risk prime upon millennial retirement investment decisions can be attenuated by objective financial literacy. Finally, Study 3 includes a qualitative study to enrich the quantitative findings and further unveil why millennials may adhere to conservative investment strategies. Specifically, this last study examines the existence of a cohort effect, as suggested by the literature discussing millennial retirement investment decisions alongside a generational cohort theory (Strauss and Howe 1991). These findings consistently illustrate that millennials have been impacted by coming of age within a tumultuous social and economic environment, while often possessing low financial knowledge. With a greater understanding of the millennial segment, marketers may be better able to address the financial perceptions of millennials and enhance their investment decisions.

The article proceeds as follows. First, the literature on millennials is covered, including discussion of potential cohort differences in response to events such as the Great Recession. Then, the literature on knowledge and risk in regard to financial decision making is discussed and hypotheses are presented. Three studies and their corresponding results are presented. Finally, our study concludes with a discussion of findings and suggestions for future research.

THEORETICAL BACKGROUND AND RESEARCH HYPOTHESES

Cohort Theory and the Millennial Generation

Wolburg and Pokrywczynski (2001) suggest three major influences found in generational marketing research: life stage, current conditions, and cohort

experiences. Generational cohort theory (Strauss and Howe 1991), which is grounded in the life course perspective (Elder 2006), proposes that major historical events and societal changes can impact individuals who are within their formative years at the time of those events, creating distinct behavioral and psychological profiles for each generation. A generational cohort is, therefore, a group of individuals with shared similar experiences and unique common characteristics around these experiences (Beldona, Nusair, and Demicco 2009). Thus, a particular cohort is associated with certain unique values and priorities that may persist over their lifetimes (Jackson, Stoel, and Brantley 2011) resulting in each cohort exhibiting distinct attitudes and behaviors (Moore and Carpenter 2008). Because cohorts have been impacted in a similar way by external events, generational cohorts may be a more effective means to segment a market than just by age (Schewe, Meredith, and Noble 2000).

Cohorts are somewhat different than generations as they are defined not by year of birth, but rather by the external events that impact the cohort during their late adolescent/early adulthood years (Schewe, Meredith, and Noble 2000). Consequently, different cohorts can be of very different lengths of time (Meredith and Schewe 1994). While there is no consensus on the exact age range, scholars tend to agree that millennials, also known as Generation Y, are defined as those born somewhere between 1977 and 2000 (Norum 2003), although they are more generally defined as being born around the early 1980s up through the early 2000s (Gloeckler 2008). The Pew Research Center further divides up the cohort into two distinct subsets: defining adult millennials as those who were born between 1981 to 1997 and young millennials as those born in 1998 and onward (Pew Research Center 2014).

It has been suggested that, as a whole, the millennial generation has different values, characteristics, and behaviors compared with previous generations (Eastman and Liu 2012; Gurau 2012; Howe and Strauss 2003), is larger than previous generations (Nowak, Thach, and Olsen 2006), and is seen by marketers as having a high level of spending power (Wolburg and Pokrywczynski 2001). Also, adult millennials (Pew Research Center 2014) are far more likely to be college educated than previous generational cohorts (Sharf 2014), and those who are college educated are expected to have the greatest impact on the U.S. economy (Desilver 2014). Thus, a clear understanding of

this generation's consumption behavior is critical (Gurau 2012).

This cohort was raised during a period of “unprecedented advances in safety...[T]hese advances helped to make the millennials the most protected generation in history” (Dickerson 2007, p. A13). As a result of being sheltered, protected, and simultaneously pressured by their parents to succeed, millennials often struggle with independent thinking, decision making, and risk taking out of fear of making mistakes (Howe and Strauss 2003). These consumers also have a high need for structure with low tolerance for ambiguity (Bell, Connell, and McMinn 2011; Dickerson 2007) and are, therefore, highly risk averse (Sharf 2014). Despite being sheltered, millennials have witnessed significant turmoil in key growing years (Dickerson 2007). This environment may significantly impact their choice of investments into adulthood (Malmendier and Nagel 2011), as events occurring throughout one's formative years are often more substantial than events occurring later in life, and cohort effects may not change with age or stage of life (Meredith and Schewe 1994). Similar to the conservative money values held by those growing up during the Depression (Schewe, Meredith, and Noble 2000), millennials may also hold especially conservative investment values (Zick, Mayer, and Glaubitz 2012) due to an environment of uncertainty during their formative years.

It is important to note that previous research on the baby boomer cohort has suggested that while studying cohort effects can be valuable, there is an opportunity to glean more specific information when looking at an even tighter age range within a single cohort (Morton 2001; Reisenwitz and Iyer 2007). Noble, Haytko, and Phillips (2009) stress the need for marketers to gain a better understanding of the antecedents and motivations for college-aged millennial consumption behavior. As investment patterns and consumer lifelong perceptions of retirement planning emerge around the early 20s (Cayla, Beers, and Arnould 2014), the focus of the current paper is on the *adult* millennials (Pew Research Center 2014), as they will soon be making their first retirement decisions and are on the cusp of entry into the professional workforce.

Thus, our theoretical foundation for examining the influence of knowledge and risk on investment decisions is cohort theory, focusing specifically on the college-aged millennial segment. Major historical and

societal events within one's formative years have been demonstrated to create a distinct generational cohort profile that may persist over a lifetime (Beldona, Nusair, and Demicco 2009; Elder 2006; Jackson, Stoel, and Brantley 2011; Moore and Carpenter 2008). As the millennial generation has witnessed significant turmoil in their formative years (Dickerson 2007), the influence of knowledge and risk on millennial retirement investment decisions may be significantly impacted. Given the financial impact of the millennial generation (Cudmore et al. 2010), it is also of theoretical and managerial interest to marketers to examine how knowledge and risk influence millennial retirement decisions at the time when perceptions of retirement planning are emerging among this cohort (Cayla, Beers, and Arnould 2014).

Knowledge

Making investment decisions can be stressful and confusing given the sheer amount of information and number of choices available (Kozup, Howlett, and Pagano 2008), and investment knowledge can have a significant impact on the quality of one's investment decisions (Hershey and Walsh 2000/2001). There are three categories of knowledge: (1) subjective knowledge (self-beliefs about one's own knowledge), or what one thinks he/she knows; (2) objective knowledge (accurate stored information), or actual domain literacy that can be tested; and (3) prior experience with the domain (Brucks 1985; Carlson et al. 2009). Per Radecki and Jaccard (1995), what an individual believes s/he knows should be some function of what s/he does know. In fact, subjective knowledge has shown to be positively correlated with objective knowledge (Brucks 1985; Carlson et al. 2009; Selnes and Gronhaur 1986), which may be attributable to the amount of confidence an individual has in his/her ability to make good decisions. Research suggests that confidence in financial decision making is positively related to objective financial knowledge, and that confidence positively impacts financial planning and financial decisions, even after controlling for demographic characteristics (Parker et al. 2012).

Objective financial knowledge has been considered in terms of financial literacy, defined as “the knowledge of basic financial investment concepts such as inflation and risk diversification and the capacity to do calculations related to interest rates” (Lusardi 2012, p. 26;

Lusardi and Mitchell 2011). Given that people with higher levels of financial literacy are more likely to invest in stocks (Van Rooij, Lusardi, and Allessie 2011) and plan well for retirement, the need to enhance levels of financial literacy is important (Lusardi and Mitchell 2008; 2011).

Altogether, research suggests that investment decisions are a function of knowledge and that both subjective knowledge (Parker et al. 2012) and objective knowledge (Masters 1989) can impact investment decisions. Thus, we hypothesize:

H1: Millennials with lower (higher) levels of subjective financial knowledge will select more (less) conservative retirement options.

H2: Millennials with lower (higher) levels of self-reported familiarity of financial planning for retirement will select more (less) conservative retirement options.

Knowledge and Risk Tolerance

In a review of behavioral decision theory, Mellers, Schwartz, and Cooke (1998) note that most modern frameworks of judgment and decision making take into account the key role of risk and emotion, moving a step beyond pure rational choice theory to understand better the factors affecting actual decision making patterns. Risk is a multidimensional construct, and a “lack of familiarity” is a core dimension of risk (Mellers, Schwartz, and Cooke 1998); therefore knowledge and risk are inherently intertwined concepts. The importance of understanding consumer perceptions of risk has been addressed over the years in the literature, as risk impacts consumer information seeking and decision making (Bettman 1973; Jacoby, Johar, and Morrin 1998; Mandrik and Bao 2005). Financial risk is one of the “predominant risk dimensions,” particularly when dealing with expensive, complex, and difficult-to-judge products (Stone and Gronhaug 1993, p. 46). Financial risk tolerance is defined as “the maximum amount of uncertainty that someone is willing to accept when making a financial decision” (Grable 2000, p. 625). Most of the work in the area of financial risk tolerance focuses on the individual’s general financial investment decisions, and studies on these decisions demonstrate that risk-tolerant individuals prefer high-risk investments, whereas risk-averse

individuals (Mandrik and Bao 2005) prefer investing in bonds and certificates of deposits (Jacobs-Lawson and Hershey 2005).

Individuals greatly vary when it comes to the amount of risk they are willing to tolerate in a given situation (Mandrik and Bao 2005). Risk tolerance and risk aversion have been linked to both contextual and environmental factors (Bateman et al. 2007), knowledge and expertise (Kraus, Malmfors, and Slovic 1992; Taylor and Dunnette 1974), and confidence in one’s ability to make good decisions (Krueger and Dickson 1994). Millennials have grown up within an environment rife with risk and uncertainty, as a cohort tend to demonstrate lower levels of financial literacy (Lusardi, Mitchell, and Curto 2010; Lusardi and Mitchell 2011), and are known to struggle with confidence in decision making (Howe and Strauss 2003). While the literature indicates that financially risk tolerant individuals make more aggressive financial plans in terms of saving efforts (Jacobs-Lawson and Hershey 2005), prior research has not examined financial planning and general risk aversion, nor has prior research exclusively focused on millennials and their choice of a retirement investment portfolio. Thus, we extend existing research by proposing

H3: Millennials with lower (higher) levels of risk aversion will select less (more) conservative retirement options.

H4: Millennials with lower (higher) levels of financial risk tolerance will select more (less) conservative retirement options.

Knowledge as a Buffer

Another consideration is the way in which risk may impact those with high and low knowledge differentially. Research suggests that the greater an individual’s knowledge of investments, the greater the willingness of that individual to take risks in the market (Masters 1989). Furthermore, the propensity to take or not to take a risk is often linked to previous training and experience, attitudes, available information, and knowledge (Kraus, Malmfors, and Slovic 1992; Taylor and Dunnette 1974). In other words, the millennial cohort may be highly attuned to concerns over financial risk; however, financial knowledge may serve as a buffer, allowing some individuals to overcome the

impact of choice under risk and the tendencies of their cohort.

Some, however, do not have enough expertise or knowledge to make an informed decision when it comes to finances, and in these cases, temporary contextual factors may influence choice. Traditionally, research on financial decision making has focused on conscious processing and rational thought. However, more recent understanding of behavioral finance incorporates findings from cognitive psychology, highlighting various irrational influences on financial decision making (Paivio 2007). For example, those with low levels of financial knowledge may be more likely to choose based on what they *feel* in the moment, utilizing those feelings as relevant information, and then attributing their decision to personal preference after the fact (Tetlock 1992). Prior studies have shown that those with high levels of knowledge within a specific domain are able to make decisions within that domain independent of temporary states, while nonexperts are more heavily influenced by temporarily activated states, contexts, and mindsets (Englich and Soder 2009; Schwarz 1990). This practice may also apply to the domain of financial knowledge. Research on human problem solving and decision making (Newell and Simon 1972) has long recognized that when an individual is solving a complex problem, information processing will vary based on task complexity, and heuristics may be employed if processing demands are too high. Those with high domain knowledge will not need to rely on mental heuristics such as “feelings-as-information” (Schwarz 1990).

Therefore, the impact of current and future economic conditions on millennials’ willingness to take financial risk may vary, based on levels of financial literacy. If consumers with lower levels of knowledge are more prone to using feelings as information, temporary economic uncertainty or stability may have a greater impact on the financial readiness and retirement decisions of millennials entering the work force. However, investment decisions made by consumers with a higher level of financial literacy may not be equivalently impacted by risk. We specifically hypothesize the following:

H5: Low financial knowledge millennials primed with risk seeking (risk avoidance) will select more risky (conservative) investment choices.

H6: Investment decisions made by high financial knowledge millennials will not be affected by a risk prime.

STUDY 1 METHODOLOGY

Study 1 was conducted to extend prior work (Masters 1989; Jacobs-Lawson and Hershey 2005; Parker et al. 2012) by examining the impact of subjective financial knowledge (H1), self-reported familiarity with financial planning (H2), general risk aversion (H3), and financial risk tolerance (H4) on millennial retirement decisions. The sample consisted of 128 college students, all of whom had already taken a microeconomics course. The sample was 60.2 percent male, with an age range of 19 to 31 years (the average age being 21.86 years old), and a majority of the sample was Caucasian (80.5 percent). The researchers deemed the sample as appropriate, as the purpose of the study was not to provide point estimates of the variables, but to test the relationships among them (Calder, Phillips, and Tybout 1981), and importantly, the mean age reflected the age consumers initially tend to form preferences regarding retirement planning decisions (Cayla, Beers, and Arnould 2014).

Procedure

After obtaining informed consent, a survey scenario was presented in which participants were asked to imagine that they have been hired after graduation. Their employer asks them to set up their retirement plan in which 5 percent of their salary would be contributed toward their retirement plan, with their employer contributing a 5 percent match (See Appendix). To strengthen validity, participants were informed that their new employer required them to select benefits, in this case a retirement plan, with Human Resources as part of their new employee paperwork. As our sample consists of students planning to graduate approximately within the next year, this scenario represented a situation that they will likely be facing soon. Participants were presented with four investment options ranging from highly conservative (100 percent guaranteed income annuity with a current 1 percent return) to highly risky (100 percent stock fund based

on the S&P 500). They were then asked to select the option of their choice (See Appendix). Also included was an option not to participate in the retirement plan and, thus, not receive the 5 percent match from the company. No respondent selected this option. A basic explanation of annuities and the S&P 500 was included at the bottom of the scenario for those who might not be familiar with this terminology.

After selecting an investment option, participants completed measures of subjective financial knowledge (tested using Flynn and Goldsmith's (1999) scale within the domain of finance), self-reported familiarity with financial planning (items taken from Jacobs-Lawson and Hershey 2005), general risk aversion (items taken from Mandarik and Bao 2005), and financial risk tolerance (items taken from Jacobs-Lawson and Hershey 2005). All the items were measured with a seven point Likert Scale (Strongly Disagree to Strongly Agree); all scales were reliable (coefficient alpha above .70) and one-dimensional. Items and descriptive statistics are provided in Table 1. Attention and realism

checks were included, followed by demographic questions.

STUDY 1 RESULTS

Attention and Realism Checks

An attention check item was embedded within the survey, which asked participants to indicate that they were, indeed, reading the survey questions by selecting a specific response. Analysis indicated that 10 participants needed to be removed from the data set, leaving 118 participants. The realism of the retirement option scenario was measured with Dabholkar's (1994, p. 106) two-item realism measure (seven point Likert scale) asking if "the situation described was realistic" (5.45 mean, 1.22 standard deviation) and whether "I had no difficulty imagining myself in this situation" (5.32 mean, 1.22 standard deviation). The realism scores were consistent with prior research (Jackson, Keith, and Burdick 1984; Loviscky, Treviño, and Jacobs

Table 1
Measurement Scales and Descriptive Statistics

Scale/Item (7-point Likert Scale)	Alpha	Mean (SD)
<i>Subjective Knowledge scale (Flynn and Goldsmith 1999)</i>	.88	
• I know pretty much about Finance.		4.78 (1.33)
• I do not feel very knowledgeable about Finance. (r)		4.59 (1.53)
• Among my circle of friends, I'm one of the "experts" on Finance.		4.36 (1.63)
• Compared to most other people, I know less about Finance. (r)		4.94 (1.35)
• When it comes to Finance, I really don't know a lot. (r)		5.11 (1.36)
<i>General Risk Aversion Scale (Mandrik and Bao 2005)</i>	.73	
• I do not feel comfortable about taking chances.		3.14 (1.31)
• I prefer situations that have foreseeable outcomes.		5.03 (1.09)
• Before I make a decision, I like to be absolutely sure how things will turn out.		4.58 (1.15)
• I avoid situations that have uncertain outcomes.		3.81 (1.26)
• I feel comfortable improving in new situations. (r)		2.57 (.91)
• I feel nervous when I have to make decisions in uncertain situations.		4.38 (1.35)
<i>Financial Risk Tolerance (Jacobs-Lawson and Hershey 2005)</i>	.78	
• I am willing to risk financial losses.		4.27 (1.53)
• I prefer investments that have higher returns even though they are riskier.		4.34 (1.23)
• The overall growth potential of a retirement investment is more important than the level of risk of the investment.		3.88 (1.30)
• I am very willing to make risky investments to ensure financial stability in retirement.		4.08 (1.37)
• As a rule, I would never choose the safest investment when planning for retirement.		3.37 (1.35)
<i>Familiarity of Financial Planning for Retirement (Jacobs-Lawson and Hershey 2005)</i>	.86	
• I am more familiar than most people about financial investments.		4.42 (1.26)
• I am more familiar than most people about retirement planning.		3.95 (1.38)
• When I have a need for financial services, I am familiar with where to obtain information on what to do.		4.75 (1.32)
• I am familiar about how Social Security works.		4.44 (1.46)
• I am familiar about how private investment plans work.		4.31 (1.39)

2007); since less than 10 percent of the sample disagreed that the scenario was realistic, the authors determined that the sample considered the scenario realistic. Thus, participants from the millennial sample could envision themselves beginning their career after graduation and having to make a retirement plan decision as part of the Human Resource paperwork.

Selected Retirement Plan and Hypothesis Testing

The sample was fairly conservative in its choices: 15.3 percent selected the 100 percent guaranteed annuity, 65.3 percent selected the 75 percent guaranteed annuity, 16.1 percent selected the 25 percent guaranteed annuity, and 3.4% selected the 100 percent S&P stock fund. The traditional rule of thumb is to have the percentage of stocks in one's retirement portfolio be 100 percent less one's age, and the new rule of thumb is to have the percentage of stock in one's retirement portfolio be 110 to 120 percent less one's age (CNN 2013). However, the majority of the sample selected retirement portfolios more appropriate for employees nearing or in retirement rather than individuals starting their working life. This breakdown is supported by other findings that millennials will tend to invest less than 40 percent of their savings in stock (Sharf 2014), and results suggest that millennials may not have enough savings to retire and keep up with inflation (Gillespie 2011; Kelly 2013).

The hypotheses were tested using ANOVA with perceived financial knowledge (H1), self-reported financial planning familiarity (H2), general risk aversion (H3), and financial risk tolerance (H4) as dependent variables

and retirement choice as an independent variable. The multivariate results indicated a significant effect of retirement choice (Wilks' Lambda = .72, $F [12, 293.97] = 3.26$, $p < .01$); therefore, the univariate results were further analyzed for hypothesis testing (see Table 2 for mean scores of knowledge and risk across retirement choice).

H1 predicts that individuals who report having lower levels of subjective financial knowledge will select more conservative retirement options compared to individuals who report having higher levels of financial knowledge. The univariate results reveal a significant relationship ($F (3, 117) = 3.56$, $p < .05$, $\eta^2 = .09$) in support of H1. Individuals with lower (higher) degrees of subjective financial knowledge selected a more conservative (risky) retirement option ($\mu_{100\% \text{ annuity}} = 4.41$ vs. $\mu_{100\% \text{ stock}} = 6.15$) (See Table 2). This finding is consistent with the literature suggesting that retirement choice is a function of knowledge (Masters 1989). Univariate results also reveal a significant relationship between financial planning familiarity and retirement choice ($F (3, 117) = 3.95$, $p < .05$, $\eta^2 = .09$) in support of H2; individuals less (more) familiar with financial planning selected more conservative (riskier) retirement options ($\mu_{100\% \text{ annuity}} = 3.87$ vs. $\mu_{100\% \text{ stock}} = 5.30$) (See Table 2). These findings are consistent with those of Jacobs-Lawson and Hershey (2005), who found a positive association between familiarity of financial planning and more aggressive saving behavior in adult populations. H3 predicts that individuals who report higher (lower) levels of general risk aversion will select more conservative (riskier) retirement options. Mean scores for general risk aversion were lowest for individuals selecting the riskiest retirement choice

Table 2
Knowledge and Risk Differences across Retirement Choices

Retirement Choice	H1: Financial Knowledge*	H2: Financial Planning & Retirement Familiarity**	H3: General Risk Aversion (MS)	H4: Financial Risk Tolerance**
100% Annuity	4.41 (1.32)	3.87 (1.15)	3.92 (0.75)	3.60 (0.94)
75% Annuity and 25% Stock	4.66 (1.12)	4.32 (1.10)	4.97 (0.73)	3.89 (0.96)
25% Annuity and 75% Stock	5.19 (1.12)	4.88 (0.66)	3.90 (0.87)	4.48 (0.84)
100% Stock	6.15 (1.23)	5.30 (1.18)	2.96 (0.88)	5.35 (0.44)

Notes:
 * $p < .05$
 ** $p \leq .01$
 MS = Marginally significant ($p = .09$)

($\mu_{100\% \text{ stock}} = 2.96$), and the univariate results indicate a marginally significant effect ($F(3, 117) = 2.26, p = .09, \eta^2 = .06$). Therefore, partial support was found for H3. Lastly, the univariate results indicate a significant relationship between financial risk tolerance and retirement choice ($F(3, 117) = 6.02, p < .01, \eta^2 = .14$). Consistent with the prediction that individuals who report higher (lower) levels of financial risk tolerance select riskier (more conservative) retirement options, risk tolerance increased as individuals selected riskier retirement options ($\mu_{100\% \text{ annuity}} = 3.60$ vs. $\mu_{100\% \text{ stock}} = 5.35$); thus, supporting H4.

In summary, the findings suggest that millennials with lower levels of subjective financial knowledge, lower familiarity with financial planning, and lower financial risk tolerance select more conservative retirement options compared to those with higher levels of financial knowledge, higher familiarity with financial planning, and higher financial risk tolerance. While risk-taking was measured in terms of self-reported financial risk tolerance and general risk aversion, a stronger association was found between financial risk tolerance and retirement investment choice than between general risk aversion and retirement investment choice. Results from Study 1 demonstrate that financial risk tolerance and financial familiarity play a core role in millennial investment decisions. Income guarantees are likely important to this generation based on the financial experiences they and their family may have faced (Insured Retirement Institute 2013; Stanley 2013), and this trend is consistent with the fear millennials may have regarding investing (Davis 2012).

Therefore, beyond these relationships, it is important to understand the effect that an environment with high risk salience may have on those with low levels of financial literacy (i.e., objective knowledge). The significance of risk is further investigated in Study 2 by assessing how a risk manipulation impacts millennials' investment decisions depending on their financial literacy levels. Research indicates that decisions made by individuals with low levels of domain knowledge are more heavily influenced by contexts and mindsets (Englich and Soder 2009; Schwarz 1990). Domain knowledge enables efficient information processing (Reyna and Lloyd 2006; Ste-Marie 1999), and those possessing knowledge are less prone to relying on mental heuristics such as "feelings-as-information" (Schwarz 1990). This evidence suggests that a primed

risk manipulation should have a more pronounced effect on the financial decisions of individuals with low levels of financial literacy compared to individuals with high levels of financial literacy. Therefore, Study 2 was conducted to examine the combined impact of knowledge (examined as financial literacy) and risk on millennial investment decisions (H5 and H6).

STUDY 2 METHODOLOGY

Two hundred, thirty-four students participated for class credit in an anonymous web-based survey platform. The sample was 56 percent male, 71 percent Caucasian, ranging from 18 to 30 years of age with a mean age of 21.25 years. Again, the sample age fit the time period in which consumers initially tend to form preferences regarding retirement planning decisions (Cayla, Beers, and Arnould 2014).

Procedure

As in Study 1, a scenario methodology was used. However, after granting informed consent, participants first engaged in a word-ranking task requiring them to categorize a list of words in order of their use within ordinary speech (from most commonly to least commonly used words). The researcher's true intention was for participants to engage with the list of words in an attempt to prime or activate associated mindsets of risk-seeking ($n = 120$) and risk-aversion ($n = 114$). A standard "multiple unrelated studies" paradigm was utilized within the task description to reduce experimental demand (Bargh and Chartrand 2000), such that the word-ordering task had seemingly no relationship to subsequent measures in the eyes of the participants. To ensure this was the case, a funneled debriefing was utilized after the study in which participants were asked to guess the purpose or hypothesis of the study. Two participants articulated a suspicion that the first task may have affected their responses; therefore, this participant data was removed, leaving 232 participants eligible for further analysis.

The priming task was adapted from Erb, Bioy, and Hilton (2002). In the *risk-seeking* condition, participants were asked to categorize a set of words associated with positive aspects of risk-taking (bold, enterprising, daring, adventurous) and negative aspects of risk-avoidance (anxious, cowardly, timid,

neurotic) along with several distractor words to keep the prime subtle (polite, orderly, conceited, colorful, deceitful, disagreeable, thankful). In the *risk-avoidance* condition, participants categorized a set of words associated with negative aspects of risk-taking (thoughtless, dangerous, risky, reckless) and positive aspects of risk-avoidance (conscientious, responsible, careful, vigilant) along with the same set of distractor words. Before running the study, the ability of this task to prime a risk-taking and risk-avoidance mindset was piloted with a small set of students ($n = 10$) who, after completing the categorization task, responded to the general risk aversion scale from Study 1 (Mandarik and Bao 2005). Within this small pilot sample, the results suggested that the priming manipulation was functional ($p < .01$).

After completing the sorting task, participants were told that they would move on to a separate study about financial decision making, and were exposed to the retirement scenario presented in Study 1. This second task was employed to determine the impact of an activated state upon retirement investment selection. After selecting an investment option, participants completed measures of objective financial knowledge and basic demographics. Unlike Study 1, we chose to measure objective knowledge because while a risk prime could possibly alter confidence strength within a self-reported measure of subjective financial knowledge, it should not affect a participant's ability to answer objective financial literacy questions correctly. Objective knowledge was measured with three financial literacy items meant to measure one's understanding of interest rates, inflation, and diversification (Lusardi and Mitchell 2007; 2008; 2011; Lusardi, Mitchell, and Curto 2010; Lusardi 2012).

STUDY 2 RESULTS

Realism Checks

Realism checks were identical to Study 1. Less than 7 percent of the sample disagreed with the statement "the scenario described was realistic" and less than 15 percent of the sample disagreed with the statement "I had no difficulty in imagining myself in this situation." These results suggest that participants considered the scenario realistic. This finding again demonstrates that the millennial sample could envision starting their career after graduation and needing to make retirement

decisions as part of the Human Resource paperwork. As Study 2 involved an experimental manipulation rather than lengthy survey items as in Study 1, attention checks were not utilized.

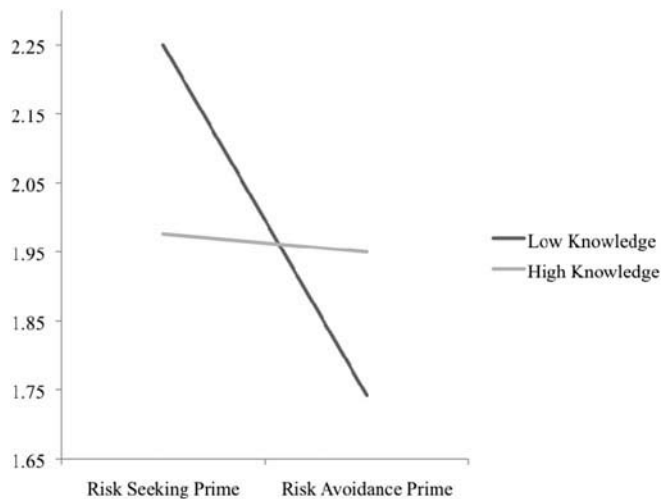
Selected Retirement Plan and Hypothesis Testing

Consistent with the findings in the prior study, participants leaned toward selecting more conservative retirement plans. 18.5 percent selected the 100 percent guaranteed annuity, 64.2 percent selected the 75 percent guaranteed annuity, 14.2 percent selected the 25 percent guaranteed annuity, 0.9 percent selected the 100 percent S&P stock fund, and 2.2 percent selected the option not to participate in the retirement plan. Participants that chose the option not to participate were excluded from further analyses (final $n = 227$). Similar to Study 1, we found the millennial sample overwhelmingly selected retirement portfolios that would not be appropriate for age and years until retirement.

Participants were classified into high and low objective knowledge groups. Those who correctly answered two to three of the objective knowledge (financial literacy) items (Lusardi and Mitchell 2007; 2008; 2011; Lusardi, Mitchell, and Curto 2010; Lusardi 2012) constituted the high knowledge group while those who correctly answered only one or none items constituted the low knowledge group. A 2 X 2 (prime condition X knowledge group) ANOVA was run on the selected retirement plan. For the analysis, the retirement plan option was considered metric, along a continuum of one to four representing the degree of stocks in one's selected retirement plan (i.e., 1 = 100 percent guaranteed annuity, 4 = 100 percent Standard & Poor's stock fund).

The results indicate no significant main effect of knowledge, $F(1, 223) = 0.14$, $p > .05$, with a significant main effect of risk prime, $F(1, 223) = 9.09$, $p < .01$; however, these main effects were qualified by a significant interaction between objective knowledge and risk condition, $F(1, 223) = 7.39$, $p < .01$ (See Figure 1). Thus, tests of simple main effects were performed to analyze further the interaction between knowledge and risk as predicted in H5 and H6. The simple main effect of risk within the low knowledge group was significant, $F(1, 223) = 11.37$, $p < .01$, which reveals that when exposed to a risk avoidance

Figure 1
Risk Prime and Knowledge Interaction



prime, low knowledge consumers select more conservative retirement plans ($\mu = 1.74$), but when exposed to a risk seeking prime, low knowledge consumers select riskier retirement plans ($\mu = 2.25$), thus, supporting H5. Likewise, consistent with H6, the simple main effect of risk within the high knowledge group was not significant, $F(1, 223) = 0.08, p > .05$; therefore, revealing that high knowledge participants are not significantly influenced by the temporary state elicited using the risk prime, such that their selected retirement plans are similar across both risk avoidant and risk seeking conditions (See Figure 1). These results are consistent with the theory (Englich and Soder 2009; Schwarz 1990) advocating that temporary internal states evoked by external stimuli may have a stronger impact on low knowledge individuals compared to high knowledge individuals.

In summary, both Study 1 and Study 2 demonstrate that millennials continually make suboptimal, overly conservative investment decisions and that these decisions are a function of knowledge and risk. To enhance our quantitative findings, a qualitative study was performed to gain a deeper understanding of millennial investment decisions. Our intention with the qualitative third study was to utilize an explanatory sequential mixed method in which a qualitative study is utilized after a quantitative study to more thoroughly explain the quantitative study results (Creswell 2014).

STUDY 3 METHODOLOGY

Procedure

Twenty-two undergraduates participated in the study for optional extra credit. Since detailed open-ended responses were required, extra credit was utilized to encourage participants to put forth the necessary time. The average age was 22 with a range of 21 to 24 years, and participants were predominantly female (77 percent) and Caucasian (68 percent). After giving informed consent, the students were asked a series of items utilizing an open-ended format within an online survey platform. First, participants were presented with the basic investment scenario from Studies 1 and 2, and then they were asked which retirement option they would select. After selecting a retirement option, participants were asked to describe, in detail, the reason for their selection. Next, summary information from Study 1 was presented, which illustrated the percentage of their peers who selected each investment option in the prior study. Participants were then asked to describe why their peers might have responded in this manner. Finally, demographic information was gathered.

STUDY 3 RESULTS

Consistent with the prior two studies, participants selected more conservative retirement plans (27 percent selected the 100 percent guaranteed annuity, 64 percent selected the 75 percent guaranteed annuity, and 9 percent selected the 25 percent guaranteed annuity). None selected the 100 percent S&P stock fund or chose not to partake in the retirement plan. Content analysis was used to analyze the textual data; two independent and trained judges coded the data. Cohen's kappa, which considers the level of rater agreement beyond chance, equated to 85 percent. Five prevailing themes provided insight to participants' own retirement selections and the selections of their peers. In order of their frequency throughout the text, the five themes include stock riskiness, lack of financial knowledge, good decision making, economic uncertainty, and recession history (see Table 3).

Stock riskiness, the most dominant theme, is the idea that stocks are considered risky. Popular words

Table 3
Key Themes in Study 3

Theme & (Occurrence) ¹	Examples
<i>Stock Riskiness</i> : (30)	• Putting money in stock is risky. • I don't keep up with current events on the stock market and I don't want to leave any of my money up to chance. • I am not very keen on investing a lot of money into a company that could potentially go bankrupt. • The stock option could be a risky move because it is uncertain. • There are [is] more risk involved in stock funds.
<i>Lack of Financial Knowledge</i> : (14)	• I think students' investment selections could be either a lack of knowledge or uncertainty about the economy. • Students might not be educated on the potential rewards that come with stock investments. • I think most students really don't understand the importance and potentially a great payoff from stock funds. • I never fully grasped the idea of stock markets because no one ever explained it to me. • I honestly do not know anything about money right now but it seemed like the best route.
<i>Good Decision Making</i> : (13)	• I chose to invest 100% in the annuity because I do not like to purchase stock based entirely on indexes such as standard and poor. • I feel as if choosing the 75% for retirement will be the best option. • The investment of stock funds is attractive if planning for retirement.
<i>Economic Uncertainty</i> : (11)	• Uncertainty with the economy is a huge factor. • Because the economy is so fragile that makes stocks very risky to invest in. • The economy can have a great effect on the returns on investments in stock funds. With a shaky economy, it is even more difficult to predict returns • Our economy the last few years has been unstable. Many people I know who have investments have lost money and not done well. I have no idea what the future is like.
<i>Recession History</i> : (7)	• History has shown them [stocks] that it is [they are] not always the best idea. • Many of the S&P 500 companies were among the many to suffer during the economic recession. • Especially with the recession and stock market crash in 2008, people are weary about the stock market. • The recent stock market crash is still fresh on everyone's mind and is a factor in determining choices in regard to the stock market.

Note: ¹Indicates the number of times a theme was mentioned throughout the twenty-two open ended responses.

mentioned by participants representing this theme included "not guaranteed," "risky," "possible loss," and "uncertainty" (see Table 3). In addition to these examples, the textual data also make it evident that participants are afraid of losing money in the stock market. Therefore, *stock riskiness* represents negative views of stock funds. Comments mentioning possible loss included statements such as "Stocks are not guaranteed. The money you invest in stock can be lost in an instant"; "I want my retirement fund to be definite. I don't want there to be a chance I could lose it"; and "You can easily lose money when you invest in stock. People don't want to risk more [a higher] percentage of their money to stock when they can save it in an account that will always be there."

Lack of financial knowledge, the next most prevalent theme, represents the notion that millennials often feel that they lack knowledge about investing or finance. Key words representative of this theme include "lack of knowledge," "uneducated," and "don't understand."

Select statements from three participants highlight the lack of financial knowledge among many millennials: "Not a lot of college students know much about the stock market"; "Unless a student's studies are concentrated in this area, few know the ins and outs of investing"; and "I personally know very little about this subject. Therefore, this survey is kind of a challenge for me. I'm sure many other students struggle to decide which answer [retirement selection] is the best as well" (see Table 3 for additional examples).

As noted above, the majority of the participants selected conservative retirement funds, and only 9 percent of participants selected the 25 percent guaranteed annuity, although conventional wisdom would suggest that a retirement plan composed of diversified stock is appropriate for twenty-something millennials starting to save for retirement. Interestingly, students believed their retirement selection to be a *good decision* for them, despite their selections being contrary to conventional financial wisdom. The statement *good decision* refers to

participants believing their retirement selection suited them and/or that their retirement selection was a good decision. Comments representing this theme included “I selected that all of my money towards retirement goes into an annuity account because I think that would be the smartest option for me”; “I felt that the plan of 100% Annuity was the best”; and “I like the idea of having 10% of my salary go into a retirement fund, while only having to pay 5% of my salary” (see Table 3 for additional examples).

Economic uncertainty, which refers to perceptions of uncertainty within the economy, was also mentioned by several participants. This theme acknowledges that millennials may lack confidence and trust in the economy. Frequently, participants associated economic uncertainty with potential loss in the stock market. Statements representing this theme included, “With the economy in constant debate, students do not want to risk too much and lose it all”; “I think uncertainty in the economy is what drives students most to select the option that I did”; and “I think so few students chose not to devote a higher percentage to stock funds because of the uncertain economy that we are experiencing right now.” These comments and the examples provided in Table 3, illustrate the association between economic uncertainty and investments.

Lastly, the Great Recession continues to be associated with millennial investment strategies. *Recession History* refers to the impact of prior economic conditions and the adverse impact of these conditions on the participant and/or his/her family. For example, one participant stated, “Many of us either have retirement plans or have family members that have retirement plans that were greatly impacted by the recession making us less likely to take the riskier option.” Another participant stated, “Most recent college graduates likely remember the ‘fallout’ their parents might have felt when the economy tanked in 2008.” These examples, combined with those in Table 3, emphasize that prior economic conditions continue to be memorable to millennials, which may influence this generational cohort’s retirement selection.

DISCUSSION AND IMPLICATIONS

Theoretical Implications

The studies herein consistently demonstrate the significance of risk and knowledge on millennial retirement

investment decisions. Triangulation (i.e., utilizing survey, experiment, and qualitative methods) permitted a deeper understanding of millennials and allowed us to exploit the advantages unique to each methodology and limit potential method bias (Creswell 2014; Jick 1979). Overall, our research extends the marketing literature by examining the constructs of subjective knowledge and risk in the domain of financial retirement investments; by illustrating the impact of feelings-as-information for millennial consumers with low levels of objective financial knowledge; and by providing evidence of a cohort effect for the millennial generation in the domain of financial retirement investments.

In Study 1, millennials chose highly conservative retirement plans, and those who indicated low degrees of subjective knowledge about finance, financial planning familiarity, and financial risk tolerance were especially prone to making conservative retirement investment choices. In Study 2, these findings were extended by examining objective knowledge along with a risk prime to further understand the combined influence of objective knowledge and risk on millennial retirement choices. As expected, investment selections made by millennials high in financial knowledge were unaffected by the prime; however, those low in financial knowledge (i.e., financial literacy) were influenced by the risk prime. This suggests that temporary internal states evoked by external stimuli have a stronger impact on low knowledge individuals compared to high knowledge individuals, as anticipated by the literature on expertise and cognitive processing (Englich and Soder 2009; Schwarz 1990). Finally, in Study 3, five key themes enriched our understanding of millennial retirement investment choices: *stock riskiness*, *lack of financial knowledge*, *good decision making*, *economic uncertainty*, and *recession history*. These qualitative findings suggest that millennials consider stocks as risky, that they are still uncertain about the economy, and that they still ponder the recession’s impact on the economy. Additionally, while millennials note their lack of financial knowledge, many still feel that their selected retirement plan was a good decision, reflecting the issue of this cohort’s misplaced optimism regarding retirement readiness (De Hauw and De Vos 2010).

Importantly, the results offer evidence of a cohort effect, and support Debevec et al.’s (2013) claim that the Great Recession is a defining moment for the college-age millennial. While the Great Recession may not

have directly affected millennials to the same degree as their parents or relatives, Mittal and Roynes (2010) suggest that the intergenerational influence on the young is strong. In support of this view, external events, arguably, have significantly impacted millennial perceptions of investing (Davis 2012). This assertion is consistent with our results indicating that the overly conservative retirement investment choices of millennials may demonstrate cohort effect, as represented in the themes *stock riskiness*, *economic uncertainty*, and *recession history*.

Societal Implications

Overall, these results suggest significant issues concerning the retirement investments of millennials, including their tendency toward making overly conservative retirement investment decisions, while still perceiving that they are making good retirement decisions. The future of financially underprepared millennial retirees presents significant implications for marketers and the economy as a whole. For example, compared to prior generations, millennials have experienced higher levels of unemployment (Davis 2012; Morton 2002), and this issue has already been linked to \$25 billion in uncollected tax revenue annually, consequently, limiting funding for social services and creating tremendous costs for taxpayers (Sennott 2014). As articulated by Sennott (2014), "If this generation continues to stumble, our economy will fail." Millennials' financial decisions have tremendous implications for both their generational cohort as well as other cohorts. Specifically, if millennials are financially unprepared for the future, they will be affecting other cohorts too. In regards to the financial markets, older generations will be affected; without younger investors in the market, older investors may find fewer buyers and lower prices when selling stocks to subsidize their retirement (Davis 2012). Millennials are also poised to shape housing demands and the real estate market. Currently, over 15 million adult millennials still live at home—over one million more than the number in this situation ten years ago (Harvard University 2014). As these millennials eventually leave home to either rent or buy, it will have a tremendous impact on the housing market. Therefore, future research may benefit by addressing the larger impact of knowledge and risk upon other uncertain markets (e.g., housing) and investment products.

Millennials do seem to recognize that they have low financial knowledge (a concern also noted by Lusardi, Mitchell, and Curto 2010), and those who are comparatively low in financial literacy are more easily influenced by temporarily activated risk mindsets. Taken together, our results suggest a generational cohort unprepared to make retirement investment decisions just as they reach the age when consumers start developing lifelong investment decision preferences (Cayla, Beers, and Arnould 2014). Therefore, consistent with extant research, the findings herein suggest that millennials have a sense of misplaced optimism—that they will have enough money at retirement (De Hauw and De Vos 2010)—this confidence may be sorely misguided.

Managerial Implications

The question remains: How can marketers motivate this generation to behave more advantageously? For financial marketers, given the literature and our results, successfully reaching millennials and impacting their retirement investment decisions may be especially difficult. Enhancing financial literacy is vital although it cannot be accomplished with a single seminar or benefit fair. Furthermore, a one-size-fits-all approach is unlikely to be successful (Lusardi and Mitchell 2007; 2008). Lusardi and Tufano (2009) recommend financial literacy training be conducted as part of employee assistance programs offered by employers so that employees can better manage their financial futures. Eccles et al. (2013) also suggest the positive impact of financial educational activities coming from employers. Others also stress the role of employers in increasing the financial literacy and retirement investments of millennials, such as selling insurance products in the work place (Stanley 2013), encouraging the opening of 401Ks, and the use of target date funds that may be easier for millennials to utilize in diversification (Davis 2012). Considering the retirement options selected by our sample, target date funds may make good sense for millennials. Other recommendations stress the role of financial advisors in guiding millennial retirement investments (Glover 2006; Zick, Mayer, and Glaubitz 2012). It is also critical to recognize and utilize the close ties millennials have with their parents (Debevec et al. 2013; Dickerson 2007) and the role parents can play in advising their child's retirement investments (Mittal and Roynes 2010). Finally, it is important to make

financial education simple, personal, tied with emotions and goals, and for financial advisors to develop long-term relationships with their millennial customers (Gillespie 2011).

Understanding financial literacy among young people is of critical importance, and millennials should not be considered one homogeneous group of consumers, but as a collection of submarkets (Debevec et al. 2013). Differences in financial knowledge are also based on race, gender, educational attainment, and family education/financial sophistication levels (Lusardi, Mitchell, and Curto 2010). Financial advisors need to recognize that there are generational differences in approaches to investments and there may be disconnects in investment concerns between millennial investors and their financial advisors who are probably of a different generation (Insured Retirement Institute 2013). To address this realization, Sharf (2014) suggests the role of technology in providing information in a manner comfortable for millennials. Finally, in addition to looking at objective financial knowledge (i.e., financial literacy), also critical is an examination of subjective financial knowledge and the role perceived risk plays in the retirement portfolio decisions made by young workers. Particularly, our findings suggest that financial literacy training needs to address the risks associated with investing and to educate consumers on the significance of inflation and its effect on required retirement funds.

Marketers must also consider the substantial size of this cohort. Millennials represent nearly 92 million consumers (Stanley 2013), accounting for roughly 15 percent of the U.S. investment market (Cudmore et al. 2010). They are expected to spend more than \$200 billion annually beginning in 2017 (Nelson 2012). While this research has focused on retirement investments for soon-to-be college graduates, marketers will also need to reevaluate product positioning, product costs, and the comprised attributes, as the combined impact of risk and knowledge likely spans beyond the realm of financial products.

LIMITATIONS AND FUTURE RESEARCH

The use of a college-enrolled student sample was employed in an attempt to assess those millennials who are on the cusp of beginning their professional careers. As noted previously, those who are college educated are expected to have the greatest impact on the U.S. economy (Desilver 2014), and the percentage

of millennials that attend college is substantial enough to capture the majority of this cohort (Sharf 2014). Utilizing a homogeneous sample does enhance internal validity and control and still allows for this demographic to be included within a larger theoretical framework (Thomas 2011). However, future work also needs to examine millennials who have not enrolled in college, as college enrollment is positively correlated with financial market participation. Furthermore, that educational attainment levels may impact attitudes towards risk has been suggested (Cole and Shastri 2009).

This research focused on the subset of adult millennials (Pew Research Center 2014), or those who are currently college-age, although the larger millennial generation includes those who are still much younger. Future research may consider the younger portion of this cohort as they near adulthood. Particularly, as the economy rebounds, the youngest members of this cohort will have experienced somewhat different formative circumstances, which may actually signal the defining border between this generation and the next. The current research was structured as an in-depth examination of the millennial cohort; therefore, the data does not allow for cross-generational comparisons. Future work may seek to compare directly the generations on similar tasks involving risk and knowledge constructs. In summary, future research could examine millennials who are not enrolled in college, as well as young millennials as they enter their 20s. It could also compare millennials to other cohorts.

Future work should also aim to understand how factors besides knowledge and risk influence investment decisions. Such factors might include psychographic aspects such as one's lifestyle, personality, or confidence. Confidence is likely to be of key interest, as it has been positively related to retirement planning and successful financial decisions (Burke 2010; Parker et al. 2012). Moreover, confidence can be a critical issue for female consumers, who often exhibit low confidence in their own financial knowledge (Maccoby and Jacklin 1974). Research is also needed to investigate the impact of other financial issues, such as one's overall level of debt, family educational level, salary, or financial sophistication (Lusardi, Mitchell, and Curto 2010). It would be especially valuable to further examine how young consumers utilize financial information and how to best present financial information (Kozup, Howlett, and Pagano

2008) for the millennial investor. Finally, risk could be examined in more detail. For example, future research could compare the experiential and analytic systems of comprehending risk (Slovic et al. 2004) to determine how each is triggered in financial decision making. In closing, this research presents new insights concerning the impact of knowledge and risk on investment decisions of millennials and highlights the importance of better understanding cohort effects.

ORCID

Jacqueline K. Eastman  <http://orcid.org/0000-0002-4006-8080>

Dora E. Bock  <http://orcid.org/0000-0001-6161-1508>

REFERENCES

- Bargh, John A., and Tanya L. Chartrand (2000), "Studying the Mind in the Middle: A Practical Guide to Priming and Automaticity Research," in *Handbook of Research Methods in Social Psychology*, Harry T. Reis and Charles M. Judd, eds., Boston, MA: Cambridge University Press, 253–85.
- Bateman, Ian, Sam Dent, Ellen Peters, Paul Slovic, and Chris Starmer (2007), "The Affect Heuristic and the Attractiveness of Simple Gambles," *Journal of Behavioral Decision Making*, 20 (October), 365–80.
- Beldona, Srikanth, Khaldoon Nusair, and Frederick Demicco (2009), "Online Travel Purchase Behavior of Generational Cohorts: A Longitudinal Study," *Journal of Hospitality Marketing & Management*, 18 (4), 406–20.
- Bell, Nancy S., J. A. Connell, and Nathan E. McMinn (2011), "How Will the Next Generation Change the Business World? A Report on a Survey," *Insights to a Changing World Journal*, 1 (March), 45–55.
- Bettman, James R. (1973), "Perceived Risk and Its Components: A Model and Empirical Test," *Journal of Marketing Research*, 10 (May), 184–90.
- Brucks, Merrie (1985), "The Effects of Product Class Knowledge on Information Search Behavior," *Journal of Consumer Research*, 12 (June), 1–16.
- Burke, Sandra J. (2010), "The Impact of Confidence in Evaluating on Discounting: A Missing Information Context Example," *Journal of Consumer Behaviour*, 9 (September/October), 333–48.
- Calder, Bobby J., Lynn W. Phillips, and Alice M. Tybout (1981), "Designing Research for Application," *Journal of Consumer Research*, 8 (September), 197–208.
- Carlson, Jay P., Leslie H. Vincent, David M. Hardesty, and William O. Bearden (2009), "Objective and Subjective Knowledge Relationships: A Quantitative Analysis of Consumer Research Findings," *Journal of Consumer Research*, 35 (February), 864–76.
- Cayla, Julien, Robin Beers, and Eric Arnould (2014), "Stories That Deliver Business Insights," *MIT Sloan Management Review*, 55 (Winter), 55–62.
- CNN (2013), "What's the Best Asset Allocation for My Age?" *CNN Money*, June 17 (available at http://money.cnn.com/retirement/guide/investing_basics.moneymag/index7.htm)
- Cole, Shawn, and Gaury K. Shastri (2009), "Smart Money: The Effect of Education, Cognitive Ability and Financial Literacy on Financial Market Participation," Harvard Business School Working Paper, No. 09–071.
- Creswell, John W. (2014), *Research Design: Qualitative, Quantitative, and Mixed Method Approaches*, 4th ed., Thousand Oaks, CA: Sage Publications, Inc.
- Cudmore, B. Andrew, John Patton, Kemble Ng, and Charles McClure (2010), "The Millennials and Money Management," *Journal of Management and Marketing Research*, 4 (1), 1–28.
- Dabholkar, Pratibha A. (1994), "Incorporating Choice into an Attitudinal Framework: Analyzing Models of Mental Comparison Processes," *Journal of Consumer Research*, 21 (June), 100–118.
- Davis, Mark (2012), "Stocks Are Less Alluring for Young Investors," *Kansas City Star*, Kansas City, Missouri, July 16 (available at www.mcclatchydc.com/2012/07/16/156306/stocks-are-less-alluring-for-young.html).
- Debevec, Kathleen, Charles D. Schewe, Thomas J. Madden, and William D. Diamond (2013), "Are Today's Millennials Splintering into a New Generational Cohort? Maybe," *Journal of Consumer Behaviour*, 12 (January/February), 20–31.
- De Hauw, Sara, and Ans De Vos (2010), "Millennials' Career Perspective and Psychological Contract Expectations: Does the Recession Lead to Lowered Expectations," *Journal of Business Psychology*, 25 (January), 293–302.
- Desilver, Drew (2014), "5 Facts about Today's College Graduates," *Pew Research Center*, May 30 (available at <http://www.pewresearch.org/fact-tank/2014/05/30/5-facts-about-todays-college-graduates/>)
- Dickerson, Darby (2007), "Risk Management and the Millennial Generation," *Campus Activities Programming*, 39 (January/February), A12–A16.
- Eastman, Jacqueline K., and Jun Liu (2012), "The Impact of Generational Cohorts on Status Consumption: An Exploratory Look at Generational Cohort and Demographics on Status Consumption," *Journal of Consumer Marketing*, 29 (2), 93–102.
- Eccles, David W., Paul Ward, Elizabeth Goldsmith, and Guler Arsal (2013), "The Relationship between Retirement Wealth and Householders' Lifetime Personal Financial and Investing Behaviors," *Journal of Consumer Affairs*, 47 (Fall), 432–64.
- Elder, Glenn H. (2006), "Life Course Perspective," in *The Blackwell Encyclopedia of Sociology*, 4th ed., George Ritzer, ed., Malden, MA: Blackwell, 2634–39.
- Englich, Birte, and Kirsten Soder (2009), "Moody Experts—How Mood and Expertise Influence Judgmental Anchoring," *Judgment and Decision Making*, 4 (February), 41–50.

- Erb, Hans-Peter, Antoine Bioy, and Denis J. Hilton (2002), "Choice Preference without Inferences: Subconscious Priming of Risk Attitudes," *Journal of Behavioral Decision Making*, 15 (July), 251–62.
- Flynn, Leisa R., and Ronald E. Goldsmith (1999), "A Short, Reliable Measure of Subjective Knowledge," *Journal of Business Research*, 46 (September), 57–66.
- Gillespie, Lisa V. (2011), "Getting Retirement Plan Buy-In From 'Gen Me' Employees," *Employment Benefit Advisor*, 20–24.
- Gloeckler, Geoff (2008), "Here Comes the Millennials," *Business Week*, 4109, 46–50.
- Glover, Hannah (2006), "Millennials: Conservative, Loyal, Affluent," *Money Management Executive*, available at <http://www.mmaexecutive.com/issues/20061022/72431-1.html>.
- Grable, John E. (2000), "Financial Risk Tolerance and Additional Factors That Affect Risk Taking in Everyday Money Matters," *Journal of Business and Psychology*, 14 (Summer), 625–30.
- Gurau, Calin (2012), "A Life-Stage Analysis of Consumer Loyalty Profile: Comparing Generation X and Millennial Consumers," *Journal of Consumer Marketing*, 29 (2), 103–13.
- Harvard University (2014), "The State of the Nation's Housing 2014," Harvard University's Joint Center for Housing Studies (available at http://www.jchs.harvard.edu/research/state_nations_housing).
- Hershey, Douglas A., and David A. Walsh (2000/2001), "Knowledge versus Experience in Financial Planning Solving Performance," *Current Psychology: Developmental, Learning, Personality, Social*, 19 (Winter), 261–91.
- Howe, Neil, and William Strauss (2003), *Millennials Go to College*, Great Falls, VA: American Association of Registrars and Admissions Officers and LifeCourse Associates.
- Insured Retirement Institute (2013), "They Really Are Different," *Insured Retirement Institute Insight*, 5 (October), B1–B3.
- Jackson Jr., Donald W., Janet E. Keith, and Richard K. Burdick (1984), "Purchasing Agents' Perceptions of Industrial Buying Center Influence: A Situational Approach," *Journal of Marketing*, 48 (Fall), 75–83.
- Jackson, Vanessa, Leslie Stoel, and Aquai Brantley (2011), "Mall Attributes and Shopping Value: Differences by Gender and Generational Cohort," *Journal of Retailing and Consumer Services*, 18 (1), 1–9.
- Jacobs-Lawson, Joy M., and Douglas A. Hershey (2005), "Influence of Future Time Perspective, Financial Knowledge, and Financial Risk Tolerance on Retirement Saving Behaviors," *Financial Services Review*, 14 (Winter), 331–44.
- Jacoby, Jacob, Gita V. Johar, and Maureen Morrin (1998), "Consumer Behavior: A Quadrennium," *Annual Review of Psychology*, 49 (February), 319–44.
- Jick, Todd D. (1979), "Mixing Qualitative and Quantitative Methods: Triangulation in Action," *Administrative Science Quarterly*, 24 (December), 602–11.
- Kelly, Sara (2013), "Danger Zone," *Plan Sponsor*, (February), 15.
- Kozup, John, Elizabeth Howlett, and Michael Pagano (2008), "The Effects of Summary Information on Consumer Perceptions of Mutual Fund Characteristics," *Journal of Consumer Affairs*, 42 (Spring), 37–59.
- Kraus, Nancy, Torbjorn Malmfors, and Paul Slovic (1992), "Intuitive Toxicology: Expert and Lay Judgments of Chemical Risks," *Risk Analysis*, 12, 215–32.
- Krueger, Norris, Jr., and Peter R. Dickson (1994), "How Believing in Ourselves Increases Risk Taking: Perceived Self-Efficacy and Opportunity Recognition," *Decision Sciences*, 25 (May), 385–400.
- Lovisky, Greg, Linda Treviño, and Rick Jacobs (2007), "Assessing Managers' Ethical Decision-making: An Objective Measure of Managerial Moral Judgment," *Journal of Business Ethics*, 73 (3), 263–85.
- Lusardi, Annamaria (2012), "Financial Literacy and Financial Decision-Making in Older Adults," *Journal of the American Society of Aging*, 36 (Summer), 25–32.
- , and Olivia S. Mitchell (2007), "Financial Literacy and Retirement Preparedness: Evidence and Implications for Financial Education," *Business Economics*, 42 (January), 35–44.
- , and — (2008), "Planning and Financial Literacy: How Do Women Fare?" *American Economic Review: Papers & Proceedings*, 98 (2), 413–17.
- , and — (2011), "Financial Literacy and Retirement Planning in the United States," *Journal of Pension Economics and Finance*, 10 (October), 509–25.
- , —, and Vilsa Curto (2010), "Financial Literacy among the Young," *Journal of Consumer Affairs*, 44 (2), 358–80.
- Lusardi, Annamaria, and Peter Tufano (2009), "Teach Workers about the Perils of Debt," *Harvard Business Review*, (November), 22–24.
- Maccoby, Eleanor E., and Carol N. Jacklin (1974), *The Psychology of Sex Differences*, Stanford, CA: Stanford University Press.
- Malmendier, Ulrike, and Stefan Nagel (2011), "Depression Babies: Do Macroeconomic Experiences Affect Risk Taking?" *Quarterly Journal of Economics*, 126 (1), 373–416.
- Mandrik, Carter A., and Yeqing Bao (2005), "Exploring the Concept and Measurement of General Risk Aversion," *Advances in Consumer Research*, 32, 531–39.
- Masters, Robert (1989), "Study Examines Investors' Risk-Taking Propensities," *Journal of Financial Planning*, 2 (3), 151–55.
- Mellers, Barbara, Alan Schwartz, and Alan Cooke (1998), "Judgment and Decision Making," *Annual Review of Psychology*, 49, 447–77.
- Meredith, Geoffrey, and Charles Schewe (1994), "The Power of Cohorts," *American Demographics*, 16 (12), 22–31.
- Mittal, Banwari, and Marla B. Roynne (2010), "Consuming as a Family: Modes of Intergenerational Influence on Young Adults," *Journal of Consumer Behaviour*, 9 (4), 239–57.
- Mittelstaedt, H. Fred (2004), "Research and Ethical Issues Related to Retirement Plans," *Journal of Business Ethics*, 52 (2), 153–59.
- Moore, Marguerite, and Jason M. Carpenter, (2008), "Intergenerational Perceptions of Market Cues among

- US Apparel Consumers," *Journal of Fashion Marketing and Management: An International Journal*, 12 (3), 323–37.
- Morton, Linda P. (2001), "Segmenting Baby Boomers," *Public Relations Quarterly*, 46 (3), 46–47.
- . (2002), "Targeting Generation Y," *Public Relations Quarterly*, 47 (2), 46–48.
- Nelson, Erin (2012), "Millennials Want to Party with Your Brand, but on Their Own Terms," *Advertisine Age*, August 2 (available at <http://adage.com/article/digitalnext/millennials-party-brand-terms/236444/>).
- Newell, Allen, and Herbert A. Simon (1972), *Human Problem Solving*, Englewood Cliffs, NJ: Prentice-Hall.
- Noble, Stephanie M., Diana L. Haytko, and Joanna Phillips (2009), "What Drives College-Age Generation Y Consumers?" *Journal of Business Research*, 62 (6), 617–28.
- Norum, Pamela S. (2003), "Examination of Generational Differences in Household Apparel Expenditures," *Family and Consumer Sciences Research Journal*, 32 (1), 52–75.
- Nowak, Linda, Liz Thach, and Janeen E. Olsen (2006), "Wowing the Millennials: Creating Brand Equity in the Wine Industry," *Journal of Product & Brand Management*, 15 (5), 316–23.
- Olson, Jinnie (2014), "Millennials: Making Room for Retirement Savings," *Benefits Quarterly*, 30 (2), 51–55.
- Paivio, Allan (2007), *Mind and Its Evolution: A Dual Coding Theoretical Approach*, Mahwah, NY: Lawrence Erlbaum Associates.
- Parker, Andrew M., Wändi B. De Bruin, Joanne Yoong, and Robert Willis (2012), "Inappropriate Confidence and Retirement Planning: Four Studies with a National Sample," *Journal of Behavioral Decision Making*, 25 (4), 382–89.
- Pew Research Center (2014), "Millennials in Adulthood," *Generations Report*, March 7 (available at http://www.pewsocialtrends.org/files/2014/03/2014-03-07_generations-report-version-for-web.pdf).
- Radecki, Carmen M., and James Jaccard (1995), "Perceptions of Knowledge, Actual Knowledge, and Information Search Behavior," *Journal of Experimental Social Psychology*, 31 (March), 107–38.
- Reisenwitz, Timothy, and Rajesh Iyer (2007), "A Comparison of Younger and Older Baby Boomers: Investigating the Viability of Cohort Segmentation," *Journal of Consumer Marketing*, 24 (4), 202–13.
- Reyna, Valerie F., and Farrell J. Lloyd (2006), "Physician Decision Making and Cardiac Risk: Effects of Knowledge, Risk Perception, Risk Tolerance and Fuzzy Processing," *Journal of Experimental Psychology: Applied*, 12 (3), 179–95.
- Ross, Stephen A., Randolph W. Westerfield, and Bradford D. Jordan (2011), *Essentials of Corporate Finance*, New York: McGraw-Hill Irwin.
- Schewe, Charles D., Geoffrey E. Meredith, and Stephanie M. Noble (2000), "Defining Moments: Segmenting by Cohorts," *Marketing Management*, 9 (3), 48–53.
- Schwarz, Norbert (1990), "Feelings as Information: Moods Influence Judgments and Processing Strategies," in *Heuristics and Biases*, Thomas Gilovich, Dale Griffin, and Daniel Kahneman, eds., Cambridge: Cambridge University Press, 534–47.
- Selnes, Fred, and Kjell Gronhaug (1986), "Subjective and Objective Measures of Product Knowledge Contrasted," *Advances in Consumer Research*, 13 (1), 67–71.
- Sennott, Charles M. (2014), "If Millennials Continue to Stumble, Our Economy Will Fail," NBC News, October 24 (available at <http://www.nbcnews.com/news/world/if-millennials-continue-stumble-our-economy-will-fail-n232916>).
- Sharf, Samantha (2014), "The Recession Generation: How Millennials Are Changing Money Management Forever," *Forbes*, July 30 (available at <http://www.forbes.com/sites/samanthasharf/2014/07/30/the-recession-generation-how-millennials-are-changing-money-management-forever/>).
- Slovic, Paul, Melissa L. Finucane, Ellen Peters, and Donald G. MacGregor (2004), "Risk as Analysis and Risk as Feelings: Some Thoughts about Affect, Reason, Risk, and Rationality," *Risk Analysis*, 24 (2), 311–22.
- Stanley, Michael K. (2013), "Why Millennials Matter and How to Attract Them," *National Underwriter Life & Health*, 117 (August), 37–40.
- Ste-Marie, Diane M. (1999), "Expert-Novice Differences in Gymnastic Judging: An Information-Processing Perspective," *Applied Cognitive Psychology*, 13 (3), 269–81.
- Stone, Robert N., and Kjell Gronhaug (1993), "Perceived Risk: Further Considerations for the Marketing Discipline," *European Journal of Marketing*, 27 (3), 39–50.
- Strauss, William, and Neil Howe (1991), *Generations: The History of America's Future, 1584 to 2069*, New York: William Morrow.
- Taylor, Ronald N., and Marvin D. Dunnette (1974), "Influence of Dogmatism, Risk-Taking Propensity, and Intelligence on Decision-Making Strategies for a Sample of Industrial Managers," *Journal of Applied Psychology*, 59 (4), 420–23.
- Tetlock, Philip E. (1992), "The Impact of Accountability on Judgment and Choice: Toward a Social Contingency Model," in *Advances in Experimental Social Psychology*, 25, Mark P. Zanna, ed., San Diego, CA: Academic Press, 331–76.
- Thomas, Rodney (2011), "When Student Samples Make Sense in Logistics Research," *Journal of Business Logistics*, 32 (3), 287–90.
- Van Rooij, Martin, Annamaria Lusardi, and Rob Alessie (2011), "Financial Literacy and Stock Market Participation," *Journal of Financial Economics*, 101 (2), 449–72.
- Vlaev, Ivo, Nick Chater, and Neil Stewart (2009), "Dimensionality of Risk Perception: Factors Affecting Consumer Understanding and Evaluation of Financial Risk," *Journal of Behavioral Finance*, 10 (3), 158–81.
- Wolburg, Joyce, and James Pokrywczynski (2001), "A Psychographic Analysis of Generation Y College Students," *Journal of Advertising Research*, 41 (5), 33–50.
- Zick, Cathleen D., Robert N. Mayer, and Kara Glaubit (2012), "The Kids Are All Right: Generational Differences in Responses to the Great Recession," *Journal of Financial Counseling and Planning*, 23 (1), 3–16.

APPENDIX

Retirement Planning Scenario

You are getting ready to graduate from University and have accepted full-time employment in your field of study at XYZ Company. Imagine it is the first day at your new job and you are filling out paperwork with Human Resources. One set of paperwork you are completing deals with setting up your retirement plan. The XYZ Company has a program where you, the employee, contribute 5% of your salary each month toward your retirement plan and the XYZ Company matches it with a 5% contribution for a total of 10% of the value of your salary going into your retirement plan.

You must select one of the following five options for your retirement plan. Which option would you select:

Option 1: 100% of your retirement plan is invested in guaranteed income annuity*. While the rate of return is currently averaging 1%, it is 100% guaranteed

Option 2: 75% of your retirement plan is invested in guaranteed income annuity and 25% of your retirement plan is invested in a stock fund based on the S&P 500**.

Option 3: 25% of your retirement plan is invested in guaranteed income annuity and 75% of your retirement plan is invested in a stock fund based on the S&P 500.

Option 4: 100% of your retirement plan is invested in a stock fund based on the S&P 500.

Option 5: You do not participate (contribute) to the retirement plan. Thus, you do not make a 5% contribution of your salary each month towards retirement and you do not receive the 5% match from XYZ Company.

*An **Annuity** is a financial product sold by financial institutions that is designed to accept and grow funds from an individual and then provide a stream of payments to the individual at a later point in time, such as during retirement years.

** The **S&P 500**, or the **Standard & Poor's 500**, is one of the most commonly followed stock market index based on the market capitalizations of 500 leading companies publicly traded in the U.S. stock market, as determined by Standard & Poor's.
